

## Background

Foraminifera are marine plankton which generate exceptional fossil records (Figure 1). As geologists we use these to reconstruct past climates. But they also respond to environmental changes through time and space and space (Schmidt et al 2004a, 2004b, 2006). The project here is using morphometric (shape) data of these foraminifers to assess patterns of these changes and links to the climate. Understanding these traits is fundamental to our ability to assess changes in these important organisms in the future (Ying et al 2024).



Figure 1: An example of a modern deep sea fossil record of foraminifer (left individual, right assemblage) – marine calcifying plankton

In the past we have analysed their changes in size through time, tracking changes between icehouses and warm periods for example or longer term responses to changes in the ocean over millions of years.

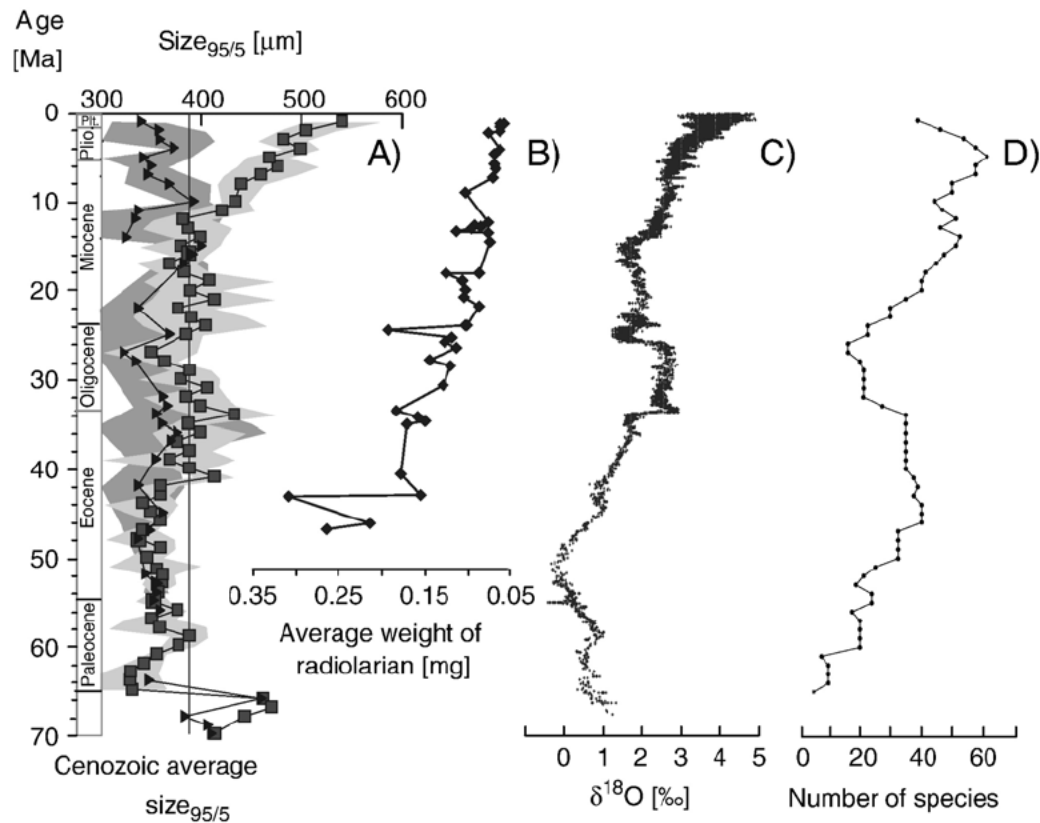


Figure 2: Changes in foraminiferal sizes over the last 66 Million years in tropical and polar waters high latitude cooling represented by  $\delta^{18}\text{O}$  ratios of benthic foraminifers (high values are cold waters low values warm), and global number of species of foraminifers (Schmidt et al 2004/2006)

But up to this point we have reduced the morphological information to assessing changes in the long tail (right skewness) of the distribution of sizes leaving many questions open.

Additional parameters we measure include area, a shape factor, elongation, sphericity, perimeter and grey value (see data files). These parameters are in the first order characteristic for a species (ie some are always smaller than others, or rounder or have a much longer periphery) and in the second order by environmental impacts on individuals.

The overarching research questions are:

- 1) Is there a change in the shape of size distribution through time not only its skewness providing more insights into changes in the community structure?
- 2) Would a more sophisticated analysis allow determining changes in number of species through time?
- 3) Are the patterns of changes through time different for size than for other parameters indicating different selection pressure?
- 4) Is there are periodicity in the data and is this different for different parameters?
- 5) Does the change in shape data link to changes in the environment.

## Data description

- Each excel files (377) represents one point in time (a geological sample from the deep sea). The excel file sample overview provides you with a full list of samples
- Each of the lines in the data represents on individual (a fossil) typically around 2000 measurements per sample. Each measurement can be linked back via the object ID to an image which we also collect.
- An overview file with all the samples labels e.g. 925C, 1H1, 43 is provided which allows to link the data to an age model (column m) and provides examples of the 95<sup>th</sup> percentile data we are calculating at the moment.
- The age model behind the age assignment can be found [here](#) in case you want to question the time series analysis. I am happy to provide you with background on how we generate age model.
- The parameters describe several morphological features and are always arranged in the same way (in case you want to automate extraction)
- Environmental data is available in this global database called Cenogrid and can be downloaded [here](#). Happy to explain this data if there is an interest to explore this.
- Large scale changes in number of taxa (morphological units) can be found [here](#). I would be very open to explain this data if of interest.